

Multiple Choice (10 marks)

1. Which of the following characteristics would not necessarily suggest that a rock we found is a meteorite.
 - (a) It has a fusion crust
 - (b) It contains solidified spherical droplets
 - (c) It is highly processed
 - (d) It has different elemental composition than earth
2. What would weigh the most on the moon?
 - (a) A kilogram of feathers
 - (b) Five pounds of bricks as measured on Earth
 - (c) Five kilograms of feathers
 - (d) A kilogram of bricks
3. Which of the following most likely explains why Venus does not have a strong magnetic field?
 - (a) Its rotation is too slow.
 - (b) It has too thick an atmosphere.
 - (c) It is too large.
 - (d) It does not have a metallic core.
4. Which living organisms most resemble the common ancestor of all life according to genetic testing?
 - (a) viruses
 - (b) bacteria such as E. coli
 - (c) organisms living deep in the oceans around seafloor volcanic vents and in hot springs
 - (d) plankton that use sunlight as an energy source through photosynthesis
5. Which is the least likely cause of death?
 - (a) Being hit in the head by a bullet.
 - (b) Being hit by a small meteorite.
 - (c) Starvation during global winter caused by a major impact.
 - (d) Driving while intoxicated without wearing seatbelts.

True / False (10 marks)

Answer True or False

6. True or False: A lunar day is roughly equivalent to 24 hours, similar to an Earth day.
7. True or False: Callisto has the smallest distance to Jupiter among the four largest moons of the planet.
8. True or False: Kirkwood gaps in the main asteroid belt occur at positions where asteroids orbit with a period half that of Jupiter.
9. True or False: When observing a star through cool hydrogen gas, the spectrum shows a blackbody spectrum with absorption lines from hydrogen.
10. True or False: The International Space Station (ISS) orbits the Earth at a speed of about 19,000 km/h, which is significantly slower than its actual speed.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the effects of rapid rotation on the shape of jovian planets, particularly focusing on the equatorial bulge and its implications for planetary formation and dynamics.
12. Discuss the implications of the angular resolution of the human eye on the observation of celestial objects, particularly when two stars are closely spaced. How would they appear to the observer?

End of Paper